

A LITTLEWOOD–RICHARDSON RULE FOR FOREST POLYNOMIALS VIA THE SCHUBERT BIALGEBRA

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ABSTRACT.

1. INTRODUCTION

2. SET-VALUED INVERSIONS TABLEAUX

Definition 2.0.1. Let w be a permutation such that $\mathbf{m}(w) \leq n$. Suppose $T : I(w) \rightarrow \mathbf{2}^{[n]} \setminus \{\emptyset\}$ is a function mapping a root to a nonempty subset of $[n]$. Define relations $\leq$ and $<$ on these sets by declaring that $A \leq B$ if $\max(A) \leq \min(B)$ and $A < B$ if $\max(A) < \min(B)$. We say that T is a *set-valued inversions tableau* if the following conditions are satisfied:

- If $i < j < k$ and $(i, j), (j, k), (i, k) \in I(w)$, then either $T(i, j) \leq T(i, k) \leq T(j, k)$ or $T(j, k) \leq T(i, k) \leq T(i, j)$.
- If $i > 0$ and $j < k$ are such that $(i, j), (i, k) \in I(w)$, then $T(i, j) < T(i, k)$.
- $T(i, i+1) \leq \{i\}$ for all i .

For $(i, j) \in I(w)$, define $\lambda_T(i, j)$ as follows. Suppose $T(i, j) = r$. Let (i, k) be the root such that $T(i, k) = r$ and k is minimal. Then $\lambda_T(i, j) = i + j + 1 - k$.

Theorem 2.1. *Suppose w is a permutation. Then we have the following formula for the double Grothendieck polynomial of w :*

$$\mathfrak{G}_w^{(\beta)}(x; y) = \sum_T \prod_{(i, j) \in I(w)} \beta^{|T(i, j)|-1} \prod_{r \in T(i, j)} x_r \oplus y_{\lambda_T(i, j)-r}$$

where the sum is over all set-valued inversions tableaux of w .

Definition 2.0.2. An *staircase word graph* is simply a subset of $\mathbb{N}^2$. We impose the same ordering on these as RC graphs. That is, we say that $(a, b) \leq (c, d)$ if $a < c$ or $a = c$ and $b > d$. We define $\mathbf{word}(G)$ to be the word obtained by reading the elements $i + j - 1$ in the linear order given by $\leq$. We then say that w_G is the Demazure product of $\mathbf{word}(G)$.

For a set-valued inversions tableau T , we can define a staircase word graph G as follows. If w is the identity, then G is the empty set. Otherwise, let n be the maximum value that is contained in $T(i, j)$ for some $(i, j) \in I(w)$. Choose i and j such that $T(i, j) = n$ and $i + j$ is as small as possible. Necessarily, $j = i + 1$, or in other words i is a right descent of w . If $|T(i, i+1)| > 1$, remove n from $T(i, i+1)$ to obtain a new function $T' : I(w) \rightarrow \mathbf{2}^{[n]} \setminus \{\emptyset\}$. If $|T(i, i+1)| = 1$, then let $T' : I(ws_i) \rightarrow \mathbf{2}^{[n]} \setminus \{\emptyset\}$ be the function defined by

$$T'(a, b) = T(s_i(a), s_i(b))$$

for all $(a, b) \in I(ws_i)$.

It is straightforward to check that T' is a set-valued inversions tableau, either for w if $|T(i, i+1)| > 1$ or for ws_n if $|T(i, i+1)| = 1$. Let $G' = \Phi(T')$. Then let $\Phi(T) = G' \cup \{(n, i+1-n)\}$.

Proposition 2.0.3. *Φ is a bijection between set-valued inversions tableaux of w with maximum value n and staircase word graphs G such that $w_G = w$ and $a \leq n$ for all $(a, b) \in G$.*

Proof. If $|T(i, j)| = 1$ for all $(i, j) \in I(w)$, then T is just an ordinary inversion tableau and $\Phi(T)$ is just the usual RC graph associated to T . In this case, the result is [1]. Otherwise, we prove this by induction on the maximum size of $|T(i, j)|$. Suppose this maximum size is $m > 1$ and we have proved the result for all set-valued inversions tableaux with maximum size less than m . We may assume without loss of generality that m is the maximum value that is contained in $T(i, j)$ for some $(i, j) \in I(w)$ by removing elements below.

In that case, $m \in T(i, i+1)$ for some i . Let T' be the set-valued inversions tableau obtained by removing m from $T(i, i+1)$ if $|T(i, i+1)| > 1$ or by applying the transformation described in the definition of Φ if $|T(i, i+1)| = 1$. By the inductive hypothesis, $\Phi(T')$ is a staircase word graph such that $w_{\Phi(T')} = w$ or $w_{\Phi(T')} = ws_i$ and $a \leq n$ for all $(a, b) \in \Phi(T')$. In either case, we have $w_{\Phi(T)} = w$. It is also clear that $a \leq n$ for all $(a, b) \in \Phi(T)$. This proves that $\Phi(T)$ is a staircase word graph with the desired properties.

Let G be a staircase word graph. Define $\Psi(G)$ to be empty if G is empty. Otherwise, let (i, j) be the maximum element of G and let $G' = G \setminus \{(i, j)\}$. Let $w' = w_{G'}$. Let $d = i + j - 1$. If $\ell(w's_d) > \ell(w')$, then let

$$\Psi(G)(a, b) = \begin{cases} \Psi(G')(s_d(a), s_d(b)) & \text{if } (a, b) \in I(w) \setminus \{(i, j)\} \\ \{d\} & \text{if } (a, b) = (i, j) \end{cases}$$

If $\ell(w's_d) < \ell(w')$, then let

$$\Psi(G)(a, b) = \begin{cases} \Psi(G')(a, b) & \text{if } (a, b) \in I(w) \setminus \{(i, j)\} \\ \Psi(G')(a, b) \cup \{d\} & \text{if } (a, b) = (i, j) \end{cases}$$

□

Definition 2.0.4. Define the isobaric divided difference operator

$$\pi^i = \partial^i(1 + \beta x_{i+1})$$

Note the Leibniz formula

$$\pi^i(fg) = (\pi^i + \beta s_i)(f)g + f\pi^i(g)$$

[2]

REFERENCES

- [1] AXELROD-FREED, I. Inversions tableaux. *arXiv preprint arXiv:2507.11516* (2025).
- [2] SAMUEL, M. J. A Molev-Sagan type formula for double Schubert polynomials. *J. Pure Appl. Algebra* 228, 7 (2024), 107636.